

XINYU SHI

Ph.D. Candidate, School of Computer Science, University of Waterloo

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Research Interests

I study **human-AI collaboration** in open-ended creative settings, aiming to *build AI systems that align with individual users' goals and values*. My work spans two connected threads: (1) **integrating human insight into AI** – surfacing the tacit knowledge, constraints, and judgment of domain experts and translating them into computational representations for AI systems [P4, P7, P9]; and (2) **designing new human-AI interaction paradigms** – constructing the common ground to externalize ambiguous human intent [P6, P8, P9] and make model behaviors inspectable [P5, P9, S3].

Education

Ph.D. in Computer Science, University of Waterloo, Canada 2021-2026 (Expected)
• Advisor: Prof. Jian Zhao

B.Eng. in Software Engineering, Xiamen University, China 2016-2020
• Advisor: Prof. Shihui Guo

Industry Experience

Adobe Research – Creative Technologies Lab Summer 2025
• Research Scientist Intern
• Hosts: Dr. Rubaiat Habib Kazi, Dr. Li-Yi Wei, Dr. Cherry Zhao
• Redefine how professional creators interact with, understand, and supervise VLMs in animation creation [P9].

Adobe Research – AI Experience Lab Summer 2024
• Research Scientist Intern
• Hosts: Dr. Shunan Guo, Dr. Jane Hoffswell, Dr. Gromit Chan, Dr. Eunyee Koh
• Modeled complex, abstract design intent into computational representations for infographic generation [S2].

Microsoft Research – Data, Knowledge & Intelligence Group Summer 2023
• Research Scientist Intern
• Host: Dr. Yun Wang
• Repurposed color visualization as the instrument to express, inspect, and iteratively refine color intent in video [P7].

Awards and Honors

Best Paper Honorable Mention Award at CHI 2026 as first author [P9] 2026
Spotlight at NeurIPS MTI-LLM Workshop as co-author [A1] 2025
Best Paper Award at CHI 2024 as first author [P7] 2024
Oral Presentation at ACM MM as co-first author for FeMaSR [P3] 2022
Jane Street Fellowship, Honorable Mention 2026
David R. Cheriton Graduate Scholarship, University of Waterloo (\$20,000 over 2 years) 2025-2027
Vector Scholarship in AI, Vector Institute (\$17,500 over 1 year) 2021-2022
International Doctoral Student Award, University of Waterloo 2021-2026
Award for Excellence in Academic Performance, Xiamen University 2016-2020

Peer-Reviewed Publications (Full Papers)

Note about venues and notions: CHI is considered as the top-tier, most prestigious venues in the field of Human-Computer Interaction (HCI), with an annual acceptance rate of 20 – 25%. 🏆 Best Paper Award; 🏅 Honorable Mention/Oral/Spotlight; * Equal Contribution.

Human-AI Collaboration & Human Intent Modeling

[P9] **Xinyu Shi**, Li-Yi Wei, Nanxuan Zhao, Jian Zhao, Rubaiat Habib Kazi. *Notational Animating: An Interactive Approach to Creating and Editing Animation Keyframes.*, ACM Conference on Human Factors in Computing Systems (CHI), 2026. **Best Paper Honorable Mention.**

- [P8] **Xinyu Shi**, Yinghou Wang, Ryan Rossi, Jian Zhao. *Brickify: Enabling Expressive Design Intent Specification through Direct Manipulation on Design Tokens.*, ACM Conference on Human Factors in Computing Systems (CHI), 2025.
- [P7] **Xinyu Shi**, Yinghou Wang, Yun Wang, Jian Zhao. *Piet: Facilitating Color Authoring for Motion Graphics Video.*, ACM Conference on Human Factors in Computing Systems (CHI), 2024. **Best Paper Award.**
- [P6] **Xinyu Shi**, Mingyu Liu, Ziqi Zhou, Ali Neshati, Ryan Rossi, Jian Zhao. *Exploring Interactive Color Palettes for Abstraction-Driven Exploratory Image Colorization.*, ACM Conference on Human Factors in Computing Systems (CHI), 2024.

Generative Model Development & Agent Evaluation

- [P5] Zijian Chen*, Xueguang Ma*, Shengyao Zhuang*, Ping Nie, Kai Zou, Andrew Liu, Joshua Green, Kshama Patel, Ruoxi Meng, Mingyi Su, Sahel Sharifymoghaddam, Yanxi Li, Haoran Hong, **Xinyu Shi**, Xuye Liu, Nandan Thakur, Crystina Zhang, Luyu Gao, Wenhui Chen, Jimmy Lin. *BrowseComp-Plus: A More Fair and Transparent Evaluation Benchmark of Deep-Research Agent.*, Annual Meeting of the Association for Computational Linguistics (ACL), 2026. **Spotlight @ NeurIPS 2025 MTI-LLM Workshop.**
- [P4] **Xinyu Shi**, Ziqi Zhou, Julia Zhang, Ali Neshati, Anjul Kumar Tyagi, Ryan Rossi, Shunan Guo, Fan Du, Jian Zhao. *De-Stijl: Facilitating Graphics Design with Interactive 2D Color Palette Recommendation.*, ACM Conference on Human Factors in Computing Systems (CHI), 2023. *Featured in Adobe News, Waterloo News, TechTimes, GraphicSpeak.*
- [P3] **Xinyu Shi***, Chaofeng Chen*, Yipeng Qin, Xiaoming Li, Xiaoguang Han, Tao Yang, Shihui Guo. *Real-World Blind Super-Resolution via Feature Matching with Implicit High-Resolution Priors.*, ACM Multimedia (MM), 2022. **Oral.**

Physical Interaction in AR/VR

- [P2] Qijia Shao, Amy Sniffen, Julien Blanchet, Megan E. Hillis, **Xinyu Shi**, Themistoklis Haris, Jason Liu, Melissa Malzkuhn, Lorna Quandt, James Mahoney, David J.M. Kraemer, Xia Zhou, Devin Balkcom. *Teaching American Sign Language in Mixed Reality.*, ACM Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT, presented in UbiComp), 2021.
- [P1] **Xinyu Shi**, Junjun Pan, Zeyong Hu, Juncong Lin, Shihui Guo, Minghong Liao, Ye Pan, Ligang Liu. *Accurate and Fast Classification of Foot Gestures for Virtual Locomotion.*, IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2019.

In Submissions

- [S3] Crystina Zhang, Zigeng Xu, **Xinyu Shi**, Raphael Tang, Nandan Thakur, Xueguang Ma, Yao Lu, Jimmy Lin. *Hawkeye: Inspecting Agentic Search Trajectories.*, (Topic: Agentic System Oversight) In submission.
- [S2] **Xinyu Shi**, Chun Ye, Shunan Guo, Jane Hoffswell, Gromit Chan, Eunye Koh, Jian Zhao. *Diagrammatic Sketching: Exploring Infographic Design Alternatives.*, (Topic: Human Intent Modeling) In submission.
- [S1] Cara Li, Xiang Li, **Xinyu Shi**, Jian Zhao. *Semantic Ink: Supporting Reuse with Variation in Annotational Graphics Authoring.*, (Topic: Human-AI Collaboration) In submission.

Academic Services

Organizing Committee — Assistant to Paper Chairs, CHI 2026

Program Committee — GI 2025, C&C 2025

Student Volunteer — GI 2026

Reviewer — UIST (2024-2026), CHI (2024-2025), TVCG (2026), VIS (2024-2025), CHI LBW (2024-2025), Pacific Graphics (2025), DIS (2024), C&C (2025). **Special Recognitions for Outstanding Reviews (×5)**

Invited Talks

Guest Lecture for CS449 at the University of Waterloo	05/2026
Invited Seminar at the CUHK (Shenzhen), School of AI	01/2026
Invited Seminar at the University of Toronto (iSchool)	11/2025
Guest Lecture for CS889 at the University of Waterloo	11/2025
UCSD Design Lab Seminar	11/2025
Invited Talk at Adobe (MOVIN Group)	09/2025